SDC Midterm01

Localization

2026/10/15-29

2026/10/20

Track03 Updates & Rule Adjustments

Dataset ➡ link

2026/10/22

Mid01 Updates Rules

link

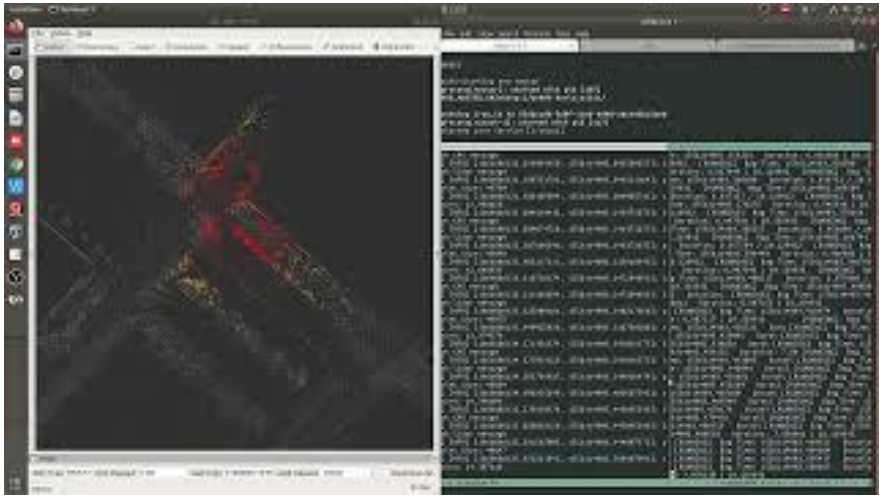
Mid01 Updates Rules 10/22

1. **No Post-processing**: Offline trajectory corrections and leaderboard-based fitting are strictly prohibited.
2. **Causal Localization Only**: Use current and past frames only. No backward or non-causal smoothing.
3. Final Submission: Select a compliant Kaggle submission by 10/29, 9:00 AM. TAs will rerun and verify the results, allowing reasonable numerical differences. Confirmed violations will result in *0 points for the affected track.
4. Code Verification: Submit reproducible code, configurations, and a README with complete execution instructions.

details are in slides

Radar / Lidar Localization

2021, 22 Lidar



2023, 24, 25 Radar

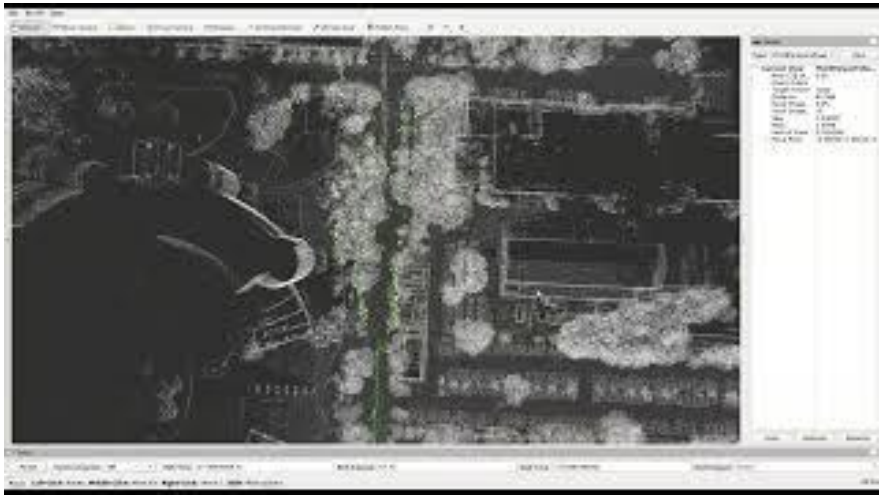
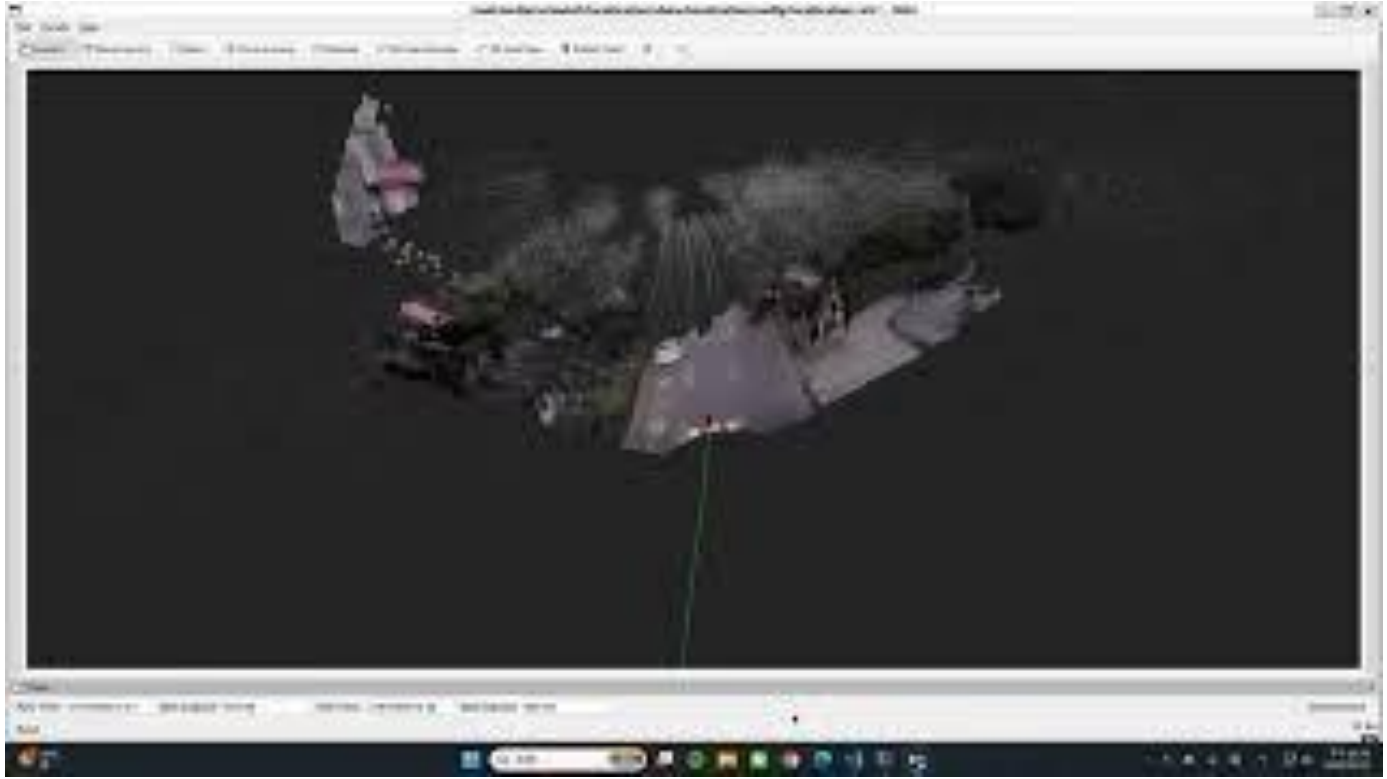
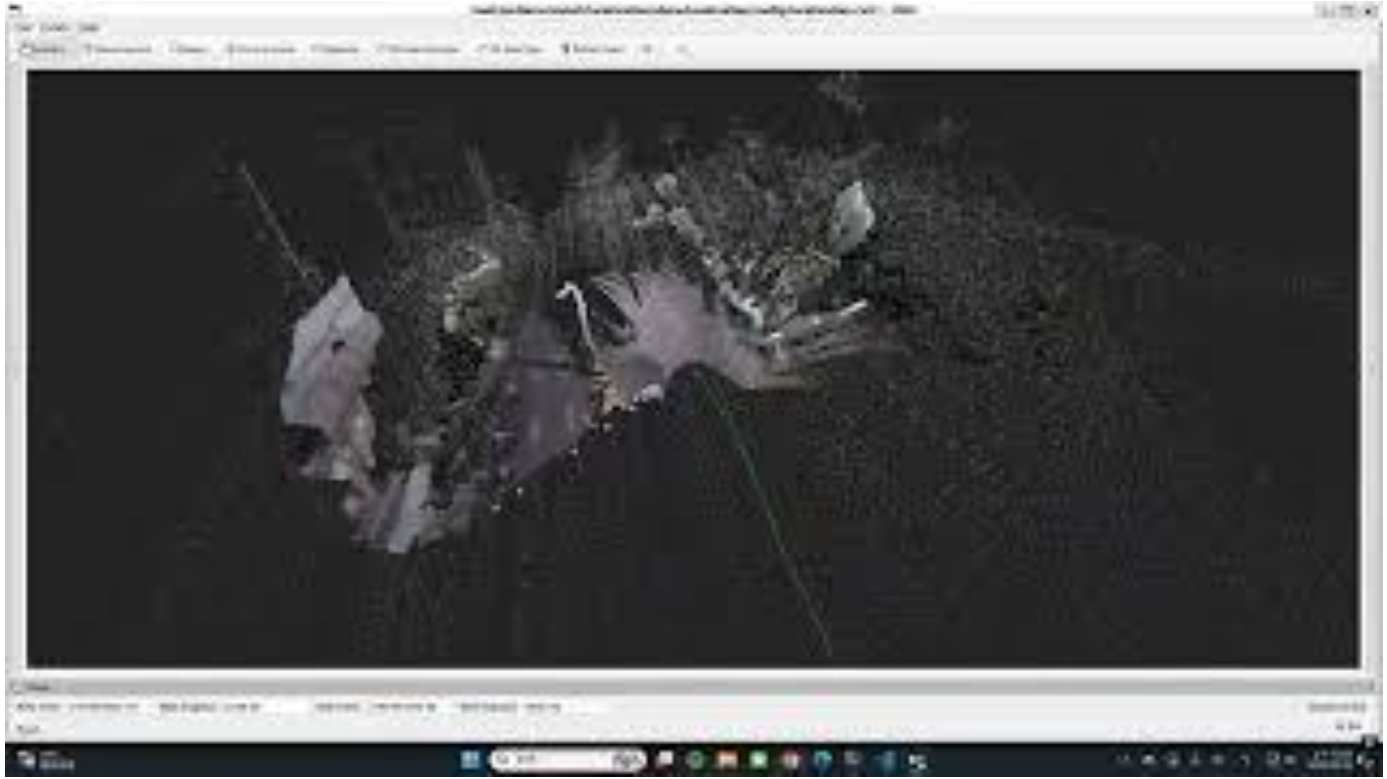


IMAGE Localization (Front-camera Version : LoGeR)



https://youtu.be/EZXgR9GIUbA

IMAGE Localization (3-Camera Multi-View Version : MapAnything)

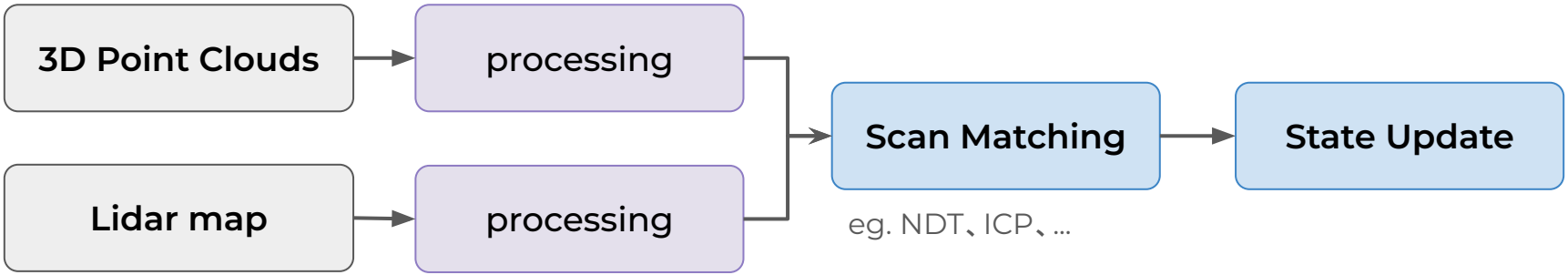


https://youtu.be/QMaRWpL5w3g

Flow



eg. Scale calibration、雜訊處理、...



eg. NDT、ICP、...

eg. 下採樣、切塊、...

Localization

Estimate the **vehicle pose** by registering the image-derived 3D point clouds to the pre-built LiDAR map using **scan matching** algorithms

- ICP
- NDT
- .etc

Dataset & Code

Dataset

There are **three tracks** and corresponding maps that you have to finish.

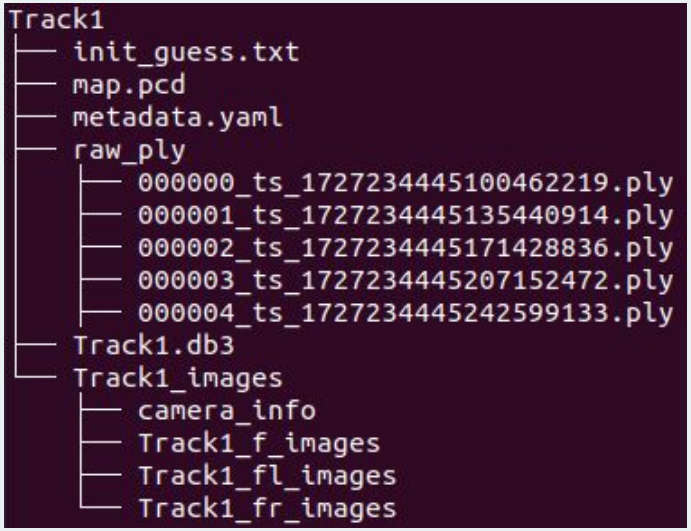
- **Track01** : 3011 image frames
- **Track02** : 3299 image frames
- **Track03** : 3142 image frames

Provided in track:

- **raw_ply** : 3D point clouds generated from front camera images using LoGeR. Named by timestamp.
- **3cameras_ply** : 3D point clouds generated from three cameras (front, front-lelf, front-right) using MapAnything.
- **Track_images** : Images extract from Track.db3, include front、front-left、front-right, 3 cameras.
- **init_guess.txt** : Initial pose with added noise.
- **map.pcd** : Pre-built 3D LiDAR map.
- **Track.db3** : ROS 2 bag containing IMU, GNSS pose, and image topics.
- **camera_info** : Contains calibration parameters for three cameras and a front-hood mask.

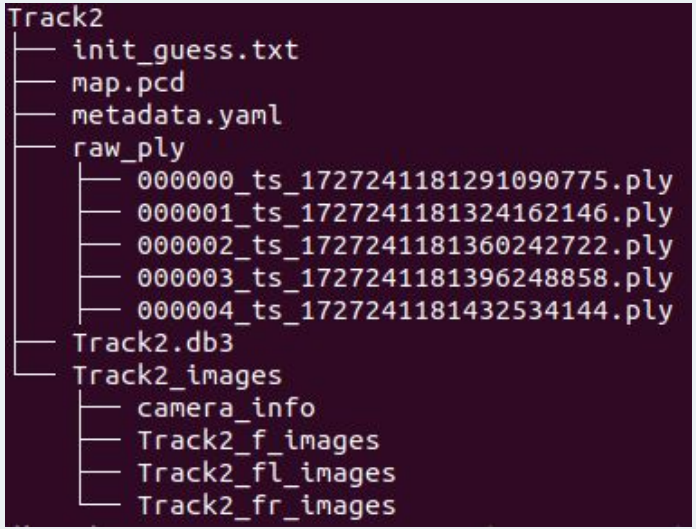
Dataset

Download from **link** (Track01)
it contains dataset



link (Track01_3cameras_ply)

Download from **link** (Track02)
it contains dataset



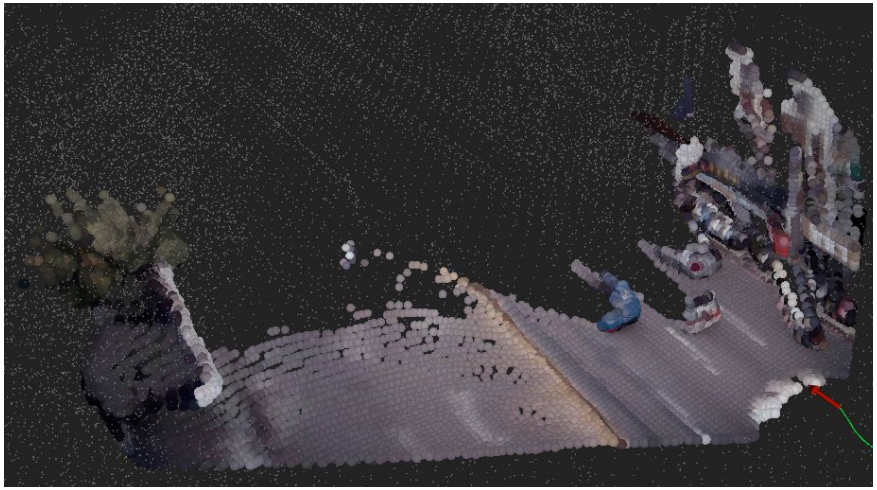
link (Track02_3cameras_ply)

TracK03 ➡ Download from **link** (Track03)

Dataset

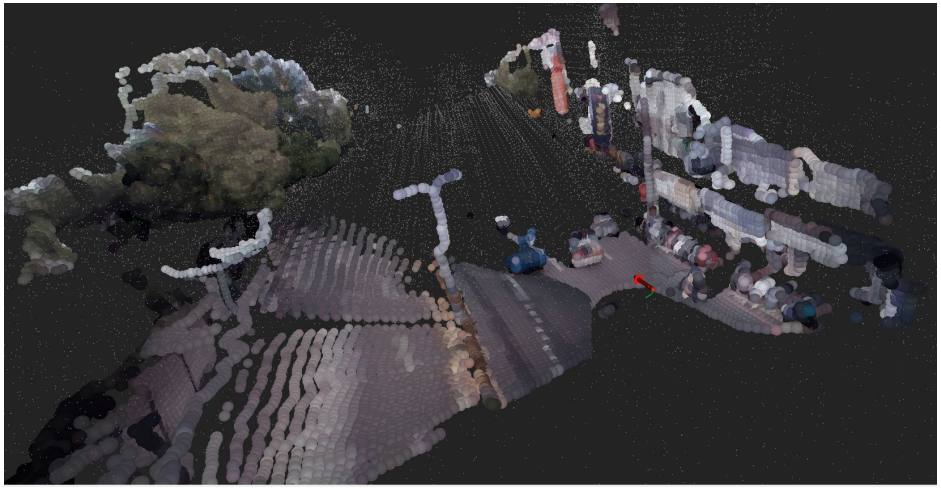
raw_ply

from front camera



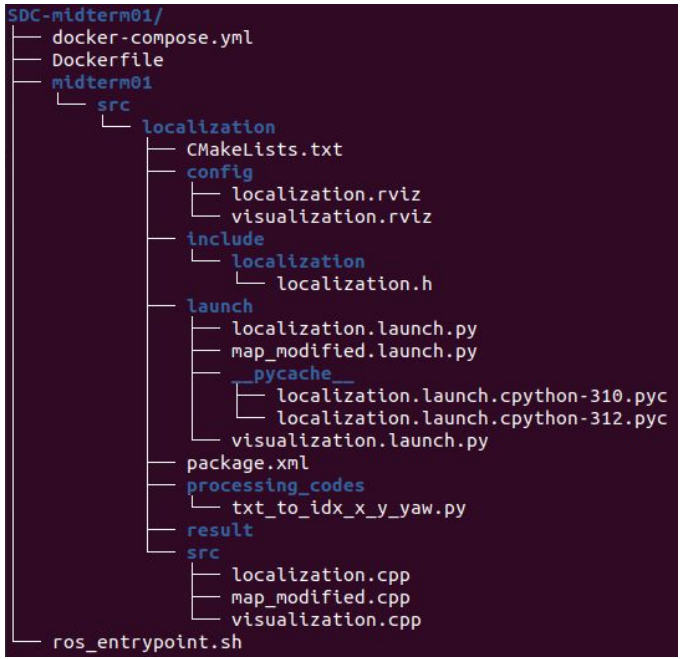
3cameras_ply

from 3 cameras, front、front-left、front-right,

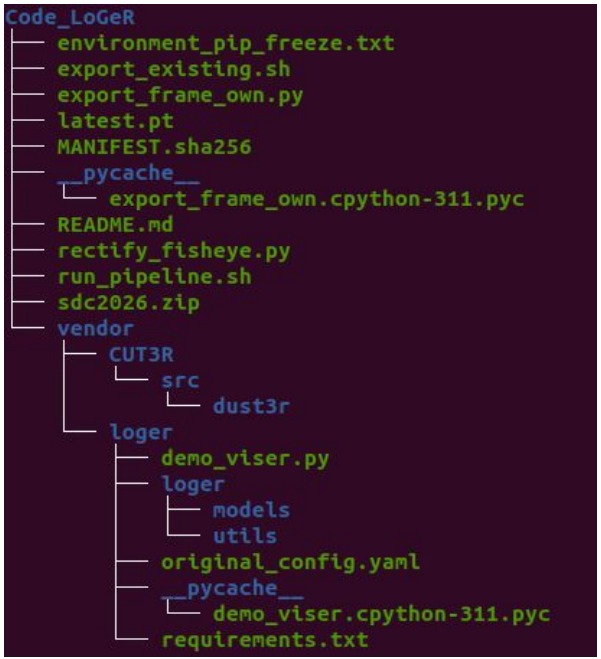


Code

Download from link (Localization)
it contains source code



Download from link (LoGeR) [optional]
it contains source code



Provided Codes

Key tasks to be completed (in /src):

- **localization.cpp** (TODO): Implement scan matching algorithm
- **map_modified.cpp** (optional): Filter/downsample map
- **visualization.cpp** : Visualize from result.txt

Depending on the situation, you may need to amend the corresponding .launch.py or .rviz file.

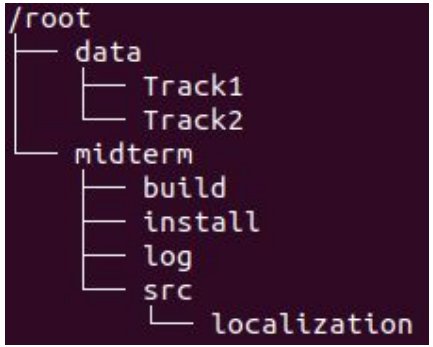
Addtional (in /process):

- **txt_to_idx_x_y_yaw.py** : Convert result.txt to Kaggle format

run the code

A Dockerfile is provided within the codebase. For detailed Docker build and run commands, please refer to **HW2**.

1. Place the provided **code** under /root/
2. Place the provided **Track1、Track2 dataset** under /root/data/




```
cd /root/midterm
colcon build
source install/setup.bash
ros2 launch localization localization.launch.py
```

Each .cpp file on the previous page has a corresponding .launch file.

Please re-run **colcon build** after modifying the source code to apply the changes.

Competition

Competition

- This is a team competition.
- Design your own scan matching algorithm
  - You can modified any template code
  - You can use other packages
  - Alternatively, you may build your solution from scratch without using the provided code
- To receive credit, your algorithm must **outperform the baseline**.
- Additional points will be awarded based on your performance ranking.

The TA baseline uses only the **front-camera-derived point clouds**. Students are free to explore additional camera images, IMU, and GNSS data to improve localization performance.

Join the competition

Use the follow links to join the competition on Kaggle

Track01 **Track02** **Track03**

- You have to submit result to kaggle
- Maximum Daily Submissions: **15**
- The result should contains estimated pose for every image frame.
- Kaggle Competition ends at **10/29, 09:00 AM**

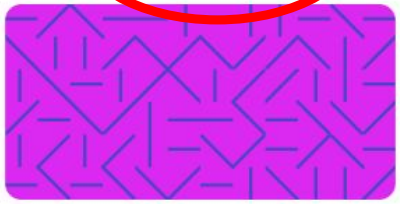
Search

NYCU_314512056 · Community Prediction Competition · 14 days to go

Join Competition

SDC2026 midterm01-Track01

Localization task, providing 3D Lidar map with image points as input



Overview Data Code Models Discussion Leaderboard Rules

Overview

Start
an hour ago

Close
14 days to go

Description

Competition Host
NYCU_314512056

Prizes & Awards
Kudos
Does not award Points or Medals

Participation
0 Entrants
0 Participants
0 Teams
0 Submissions

create team named as **group_id** and invite your teammate



SDC2026 midterm01-Track01

Localization task, providing 3D Lidar map with image points as input

Overview Data Code Models Discussion Leaderboard Rules Team Submissions

Your Team

Everyone that competes in a Competiton does so as a team - even if you're competing by yourself. Learn more.

General

Team Name

2

This name will appear on your team's leaderboard position. 1 / 30

Score Matrix-RMSE of ATE(Absolute Trajectory Error)

Δp : translation Error (in meter)

Δθ : Rotation Error (in degree)

k : weight = 0.2

$$E_n = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\|\Delta \mathbf{p}_i\|^2} + k \cdot \sqrt{\frac{1}{n}\sum_{i=1}^{n}\|\Delta \theta_i\|^2}$$

Grading

Competition Ranking : up to 56 points ⟶ **Updated Ranking Score**

Report (TA) : 30 points

Presentation (Bob) : 20 points

For each Track

- 1st : 56 ***r**
- 2nd : 52 ***r**
- 3rd : 48 ***r**
- Top 50% : 42 ***r**
- Top 75% : 36 ***r**
- Others above baseline : 30 ***r**
- Late Submission : 22 ***r** (Deadline : 11/05, 23:59 PM)
- Below baseline : 0

For Track 1, 2, **r=0.3**. For Track 3, **r=0.4**

The total score is the sum of 3 Track ranking score.

Report and Presentation

Include:

- Pipeline (How does your program work?)
- **Contributions** (Highlight your original ideas and improvements.)
- Problems and Solutions
- **Workload Distribution** (Specify each member's tasks and contribution percentage.)
- Others

Contributions are evaluated based on originality and problem-solving. Using an open-source project without meaningful modifications or original contributions will result in zero points for this component.

Prepare a 4-6 minnte presentation

- **Presentation Date : 10/29 in class**
- Introduction your work and contribution
- Email your presentation slides or slides link to neal516.en14@nycu.edu.tw by **10/29 9:00**.

E3 Submission

Every team should upload your codes(excluding data) and report to E3.

- **<student ID1>_<student ID2>_localization.zip**
  - Whole **localization/** folder, excluding the original dataset.
  - Any code you wrote
  - Your best submission files of the 3 tracks named with **result_1.csv, result_2.csv, result_3.csv**
  - Report named with **localization_report_<student ID1>_<student ID2>.pdf** ([optional]could also attach presentation video link here)
  - A **README** file describes how to run your code

Deadline: 11/05, 23:59

Track03 Updates &
Competition Rule Adjustments

2026/10/20

1. GNSS Data and Localization Methods

In Track01 & 02, students were allowed to use **GNSS** data for sensor fusion. GNSS can provide an additional **pose prior** to improve the initial guess, helping scan matching algorithms such as NDT converge to the correct pose and improve localization accuracy. This is a valid and technically meaningful localization approach that complies with the original competition rules.

However, after further discussion, we believe the primary objective of this competition is for students to estimate the vehicle pose by **registering point clouds against a pre-built LiDAR map.**

Providing GNSS poses introduces additional global localization information, which may cause competition performance to depend more on GNSS accuracy than on the effectiveness of point cloud registration itself.

2. LiDAR Map Timestamp Issue

We discovered that the **tl and th timestamp-related fields** were unintentionally retained in the **Track02 LiDAR map** due to an oversight during data preprocessing. These fields should not have been included in the officially provided map data.

These fields may contain LiDAR point observation timestamps, potentially revealing additional information about the mapping process and vehicle trajectory. However, the reliability of this information and its actual impact on localization performance have not been verified. Students are advised to use this information with caution, as it should not be considered a reliable localization reference and does not necessarily improve localization accuracy.

3. Leaderboard Probing and Public/Private Scores

Some students have proposed using **Kaggle leaderboard feedback** from multiple submissions to estimate hidden localization errors by systematically adjusting trajectory positions or orientations, and then correcting their predictions.

While standard parameter tuning and model comparison are allowed, this type of **leaderboard-based fitting** does not directly improve the localization algorithm. Instead, it uses leaderboard scores to indirectly infer hidden ground truth information. Therefore, such corrections **should not be used for official grading or ranking**.

4. Track03

To better evaluate point cloud localization performance, we have decided to introduce Track03.

1. **Provided Data**: Front, Front-left, and Front-right camera images. 3D point clouds (PLY) generated from front-camera images. Pre-built LiDAR Map
   Students may directly use the provided point clouds for scan matching or explore multi-camera images to improve point cloud quality and develop alternative localization methods.
2. Track03 will **not provide GNSS pose data**.
3. Public/**Private** Leaderboard (Private Baseline Score: 1.40)
   Public Score: Evaluates only the first 5–40% of the trajectory.
   Private Score: Evaluates the remaining portion and is used for final grading and ranking. Private scores and final rankings will be revealed after the competition closes on October 29 at 9:00 AM.
   To receive the basic score, the final Private Score must be ≤ 1.40.

Track03 is scheduled for release during class on October 22, with the same deadline as Track01 and Track02. The schedule may be adjusted if necessary.

These changes aim to reduce reliance on external localization priors and leaderboard-based fitting, allowing final rankings to better reflect localization accuracy and generalization.

Groups

Groups of no more than two people; enter your details on the left.

If you require a random group, enter your details on the right.

If you are in a group of your own, mark an ‘X’ next to it.

Deadline: 10/16, 15:00

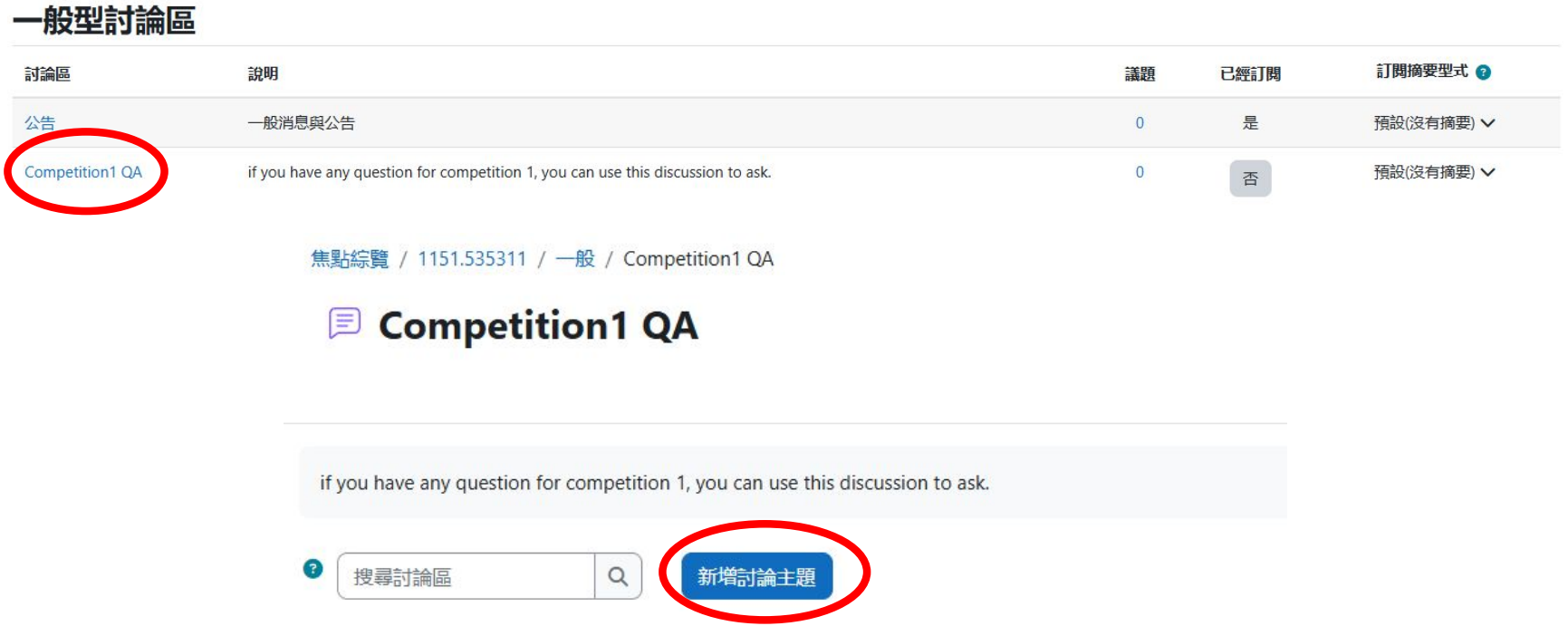
LINK

Note:
For competitions 2 and 3, you may not be in the same group as anyone you have previously been in a group with.



QA

For any questions about Competition1, please use the **Competition1 QA** discussion forum on E3.



Supplementary information

ALGO:
Class Template: ICP NDT ; 介紹: NDT

Imu:
sensor_msg/imu
Quaternion

LoGeR
MapAnything